

Adam K. Cheong



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Inventive high school senior with a strong interest in biomechanical engineering and robotics; founding intern of a venture-backed robotics startup in San Francisco; winner of multiple STEM awards and a Masason Scholar.

EDUCATION

2023–2027 Proof School, San Francisco, CA

AWARDS

Sept 2026 National Merit Commended Scholar
July 2026 AP Scholar with Distinction
March 2026 Honorable Merit, USA Biolympiad
Dec 2025 AIME Qualifier
Sept 2025 Ellison Scholar (Semi-Finalist)
July 2025 Masason Scholar
April 2025 Cum Laude, National Latin Examination (Honorable Mention)
March 2025 Honorable Mention, USA Biolympiad
April 2024 Maxima Cum Laude, National Latin Examination (Silver Recognition)

STEM FAIRS

Nov 2026 Regeneron Science Talent Search 2027
Nov 2026 International Genetically Engineered Machine (iGEM) Competition
March 2025 Fourth Place, Computational Biology. High School Division, Alameda County Science & Engineering Fair (ACSEF)
April 2024 Honorable Mention, Zoology. Senior Division, California Science & Engineering Fair (CSEF)
March 2024 First Place, Biological Sciences. 9th Grade, Golden Gate STEM Fair (GGSF)
April 2023 Honorable Mention, Applied Mechanics & Structures. Junior Division, California Science & Engineering Fair (CSEF)
March 2023 First Place, Engineering: Electrical & Mechanical. 8th Grade, Golden Gate STEM Fair (GGSF)
March 2023 First Place, Engineering, Technology & Application of Science. 8th Grade, San Mateo County STEM Fair

EXPERIENCE

Watney Robotics, San Francisco, CA (2023–present)

- Built server rack peripherals (using Onshape CAD)
- Assembled robot arm controller; trained ML models

VOLUNTEERING

- Student Ambassador, Berkeley Pre-College ('26)
- Student Volunteer, 16th Gathering 4 Gartner ('26)

SOCIETIES & CLUBS

- BioCurious Lab / KLS iGEM Team (2026–present)
- Biomedical Engineering Society (2026–present)
- Genetics Society of America (2025–present)
- Dogpatch / Benchmark Climbing (2023–present)
- 24 Hour Fitness (2024–present)
- Clock Tower Music (2023–2025)
- Berkeley Math Circle (2015–2025)

SUMMER COURSES

- DATA C8, Berkeley Pre-College Scholar (Summer '26)
- CS 61BL, Berkeley Pre-College Scholar (Summer '26)
- CS 61A, Berkeley Pre-College Scholar (Summer '25)
- Game Theory, Johns Hopkins CTY (Summer '24)
- Engr. & Robotics Workshop, MIT (Summer '24)
- Logic, Johns Hopkins CTY @ UCSC (Summer '23)

QUALIFICATIONS

- **Skills:** Use Onshape CAD to build server rack peripherals, prototype wire harness of robot controllers, assemble robot arm controllers, solder circuit boards, assemble PC, train ML models.
- **Languages:** Latin (2 years), Japanese (limited phrases), Mandarin (limited phrases), Spanish (conversational, 6 years), English (native speaker).
- **Assessments:** SAT 1550 (760 EBRW, 790 Math).

CERTIFICATIONS

- PADI Advanced Open Water Diver certification.
- Biosafety Level 1 (BSL-1) safety training.
- CPR, AED, and basic first aid certifications.

POSTERS

- Oct 2026 How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions.
2026 Biomedical Engineering Society Annual Meeting, Orlando, Florida.
- June 2026 How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions.
2026 Population, Evolutionary & Quantitative Genetics Conference,
Asilomar Conference Grounds, California.
- June 2025 Monte Carlo Simulation of Somatic Twist in Ancient Marine Worms.
25th International Worm Meeting, UC Davis, California.
- Feb 2025 A Twist on Decussation.
Junior Science & Humanities Symposium, San Francisco State University, California.

PAPERS

- Nov 2026 How to Train Your C. Elegans: Inducing Thermotaxis Behavior in a Digitally Simulated Worm.
Prepared for Regeneron Science Talent Search 2027.
- Feb 2025 Why Do Vertebrates Have Decussated Corticospinal Tracts?
Submitted to Junior Science & Humanities Symposium, NorCal, 2025.

PROJECTS

- Aug–Dec 2026 An OpenWorm simulation of *C. elegans*, with all of its 302 biological neurons in the connectome surgically replaced by artificial neurons, trained to faithfully reproduce biomechanical behavior such as thermotaxis.
- Jan–Nov 2026 A team project participated by students from 9 Bay Area high schools for the iGEM competition to engineer a biosensor that responds to rising level of cadaverine, an organic compound from decomposing meat, by producing a blue chromoprotein.
- Jan–Apr 2026 A proposed study (in collaboration with two student researchers) of cancer cells population dynamics driven by tumor growth vs. dose response within a press-pulse framework but with the added complexity of somatic mutation across treatment cycles.
- Jan–Apr 2025 Monte Carlo simulation attempt to recreate evolutionary events inside a computer and see how the first digital worm evolves *in silico* to become the ‘first fish’ with a dorsoventrally inverted body plan after a somatic twist.
- Jan–Mar 2024 Why do vertebrates have decussated corticospinal tracts? A comparative study of morphological models from ancient marine worms to the first fish inside an aquarium.
- Jan–Mar 2023 A novel mountain climbing aid that brings a new mechanical advantage to mountaineers by means of a simple lever mechanism — inspired by a physics demo harnessing a setup of three toothpicks to carry a payload hanging off the edge of a table.

REFERENCES

Dr. Michael Yetman
Director of Science Programs
Proof School

Ryan Gannon
Co-founder and CEO
Watney Robotics
